

From analysis to model: a cascade for point-process inference

1 The pitch

- AISTATS is an ML venue: a **new model** that solves a problem is a stronger paper than an analysis of where existing models fail
- This does not discard the previous work, it **uses** it: the regime analysis in **results/** identified the failure mode, and the new model is designed around it
- One-line summary: *first decide whether there is any structure to infer, and only then infer it, with every downstream model trained on data where the structure is actually visible*

2 What the exploration showed

The previous runs (`docs/05_results`) compared 22 feature sets (classical summary functions, persistent homology) on 5-way classification and parameter estimation. The consistent finding:

- **Everything breaks in the same place: near complete spatial randomness (CSR).**
 - Below $\tilde{\delta} \approx 0.6$ every method fails on Thomas (recall 0.03–0.19, below chance), calling 56–71% of those patterns Poisson
 - The lowest- $\tilde{\delta}$ bins are below chance for every feature set; the modal prediction there is Poisson
 - Parameter errors are largest near CSR and shrink as the structure strengthens (Thomas log-RMSE $1.22 \rightarrow 0.46$, LGCP $0.89 \rightarrow 0.30$ across the $\tilde{\delta}$ range)
- **No feature set fixes it.** Classical and topological inputs fail together, so the limit is information in the pattern, not the model
- **Conclusion:** don't fight the near-CSR regime — route around it

3 Design philosophy

stage	question	trained on
1. filter	Poisson, clustered or repulsive?	all simulated clouds
2. classifier	which clustered family? (Thomas, nested, LGCP)	clustered clouds <i>past the cutoff</i>
3. estimators	the family's parameters (one model per family)	that family's clouds <i>past the cutoff</i>

- A Poisson call needs no model: the intensity estimate $\hat{n} = n$ is the exact MLE. The repulsive group has one family (Matérn II), so it needs no classifier
- **Each component is trained on its own.** No component sees another's predictions during training; they are only assembled at the end
- **The whole pipeline is scored end to end** (planned): simulate from the fitted model and compare with the observed pattern by a Wasserstein distance. This rewards a *useful* fit, not a correct label, and does not reuse the summary functions the models were trained on (avoids a circular evaluation)

Why filter? (the argument to defend)

- **The filter will misfile weak structure as Poisson.** A barely-clustered Thomas pattern gets a Poisson fit
- **That is acceptable by construction:** if the structure is too weak for any method to see, a Poisson model generates patterns that look like the observed one. The error is in the label, not in the usefulness of the fit (this is exactly what the end-to-end score is designed to check)
- **In exchange, every downstream model trains on clean data:** patterns where the structure is visible, instead of spending capacity on near-CSR noise
- **First evidence that it works** (clustered classifier, on the patterns the filter actually passes to it):
 - trained on all clusters: Thomas recall 0.33 — it learns that “Thomas looks weakly clustered”, but the filter only passes the clearly clustered ones
 - trained past the cutoff: Thomas recall 0.54, overall 0.612 $\rightarrow$ 0.625

4 Stage 1: the filter

- Three very different models tie on the same 120k test patterns, which suggests a ceiling set by the data rather than the model:

model	balanced accuracy
network on summary curves (L, F, G, J)	0.796
gradient-boosted trees on summary features (chosen)	0.796
logistic regression on summary features	0.794
persistence-diagram network (PersLay)	0.774

- Chosen: gradient-boosted trees — tied for best, CPU-only, deterministic
- The remaining errors are almost all near-CSR patterns called Poisson, i.e. the regime the filter is meant to absorb

5 Where to cut: the regime analysis

Principle: measure the cutoff, don’t impose it

- For every simulated pattern we know its true parameters and whether the filter sent it to its own group
- We ask: which **coordinate of the parameters** best explains that decision?
- Candidates, per family: raw parameters, physically meaningful ones, a rough signal-to-noise estimate, and $\tilde{\delta}$ (the calibrated departure-from-CSR measure from the earlier work)
- Scored by how much of the explainable variation each captures (1 = as good as knowing every parameter); fitted on held-out predictions from the training set, so the test set never chooses the cutoff

explained.png

Result: physical coordinates explain the filter at least as well as $\tilde{\delta}$

family	coordinate	explained	$\tilde{\delta}$ explained
Thomas	cluster overlap $\omega = \sigma\sqrt{\kappa}$, corrected for $\bar{n}$	0.975	0.861
nested	inner-cluster overlap, corrected for $\bar{n}$	0.983	0.845
Matérn II	hard-core size $\times \bar{n}$	0.992	0.991
LGCP	field strength $\times$ correlation range $\times \bar{n}$	0.965	0.958

- **Interpretable:** whether a Thomas process can be detected depends on how much the clusters *overlap*, not on how many points each has
- **Two-dimensional:** more points make the same overlap easier to see, so each cutoff is on (coordinate, $\bar{n}$), fitted as one combined number coordinate $\times \bar{n}^a$

- **Honest:** $\tilde{\delta}$ is computed from the L -function, which the filter also sees, so it has a built-in advantage; physical coordinates that match or beat it are the cleaner choice

detection.png

Left: raw coordinate, curves split by $\bar{n}$. Middle: the $\bar{n}$ -corrected coordinate the cutoff uses — the curves collapse onto one. Right: $\tilde{\delta}$ for comparison.

The cutoff is a property of the problem, not of the filter



stability.png

- Trees, logistic regression and the network give the same detection curves

From curve to training data

- A pattern is “past the cutoff” at threshold τ if the filter sends patterns like it to their own group with probability $\geq \tau$
- τ is swept over $\{0, 0.3, 0.5, 0.7, 0.9\}$; $\tau = 0$ is no filtering, the baseline the filter must beat
- Higher τ = cleaner but fewer training patterns (at $\tau = 0.5$: Thomas keeps 49%, nested 83%, Matérn II 88%, LGCP 66%)

6 Status and open questions

- **Running:** every stage-2 and stage-3 model at every τ , with trees (done) and the network (in progress), then the assembled pipeline
- **Early result:** a low cutoff ($\tau = 0.3$) helps the clustered classifier most; stricter cutoffs lose too much data

- **Thomas vs nested:** their main confusion is identifiability, not features — a nested process whose outer level is smeared out *is* a Thomas process. Candidate fix: call a pattern nested only when its outer level is visible (the same filter idea, one level down)
- **Impostors:** about 5% of patterns reaching the clustered classifier are not clustered (mostly Poisson). There is no reject option yet; whether that matters is for the end-to-end score to decide
- **End-to-end score:** the first Wasserstein version could not separate the true model from CSR; it is being reworked before it is used to rank pipelines